Regression Model Building Analysis for Human Resources Survey

Info. 589-Business Statistics

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Title Page

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Introduction

This report analyzes 120 employees' data collected via a Human Resources survey. It aims to uncover trends in compensation, job level, work experience, and diversity using exploratory data analysis and regression modeling. The dataset encompasses both numerical (Salary, Age, Years Worked, Job Grade Classification) and categorical (Ethnicity) variables, making it suitable for a multifaceted analysis of equity and decision-making within the organization.

Exploratory Data Analysis Section

1. Dataset Overview

The dataset comprises 120 records across six variables:

- **Numerical**: Salary, Age, Years Worked (YrsWork), Job Grade Classification (JGClass)

- **Categorical**: Ethnicity_LABEL ("Minority" or "Not Minority"), EthnicityCODE (0=Not Minority, 1=Minority)

This forms a strong basis for analyzing workforce composition, pay distribution, and diversity.

2. Summary Statistics:

To understand the distribution of employee traits, descriptive statistics have been computed for each numerical variable. Age, Years of Work, JGClass, and Salary all show varying ranges and variations. The average wage is roughly $53,911, and most employees fall into mid-level job categories. The workforce is made up of workers of many ages and experience levels. The inclusion of 38 minority and 82 non-minority employees in the data set provides a basis for further demographic analysis.

Key summary statistics for numerical variables:

Variable	Mean	Median	Std Dev	Min	Max	Q1 (25%)	Q3 (75%)
Salary	53910.83	53100	11928.55	31200	82300	45375	62125
Age	47	49	12.95	19	74	37	57
YrsWork	6.47	6	3.99	1	18	3	9
JGClass	6.62	7	2.14	3	11	5	7.5

Ethnicity Table	
Ethnicity_Level	Count
Minority	38
Not-Minority	82

- The workforce is experienced and spans a broad range.

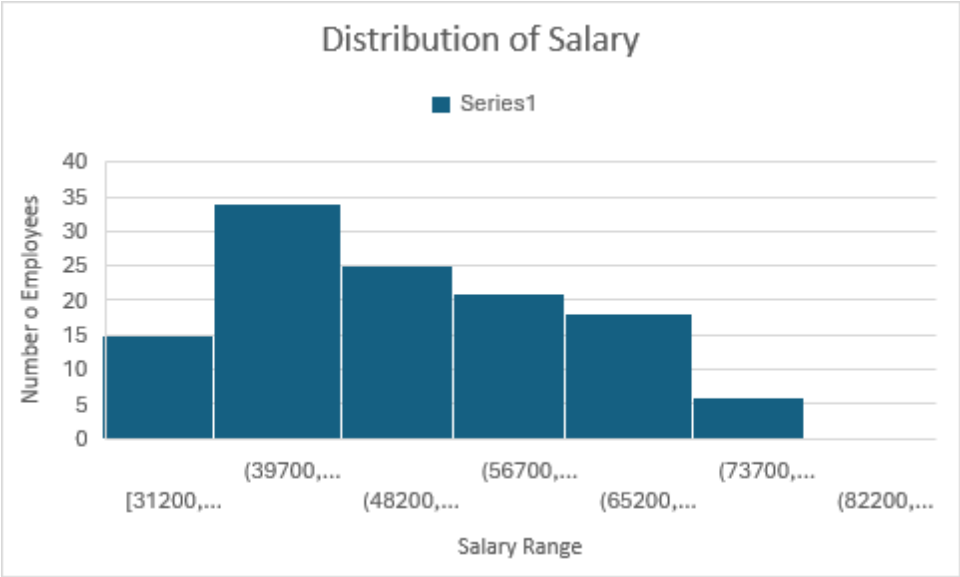
- 82 are Not Minority and 38 are Minority.

3. Histograms: Histogram presents the distribution of salary,age,years worked and job grade classification

Visualizing Distributions

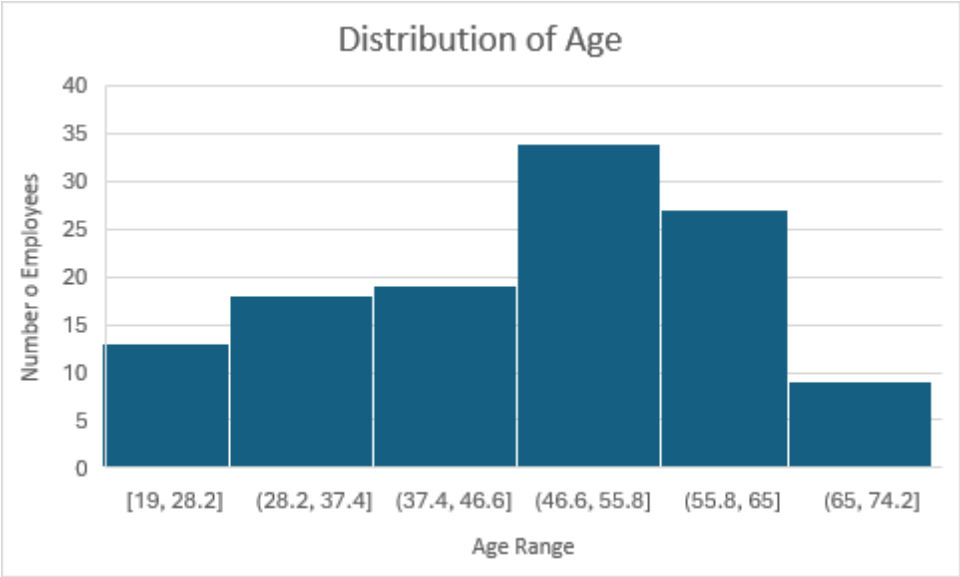
Salary: Right skewed, with the majority earning between $45,000 and $62,000. Good number of employees earn a good amount and a small number of employees earn significantly more due to the right-skewed income distribution. The range is broad, ranging from $31,200 to $82,300, with a center trend of around $53,000. This skewness highlights the differences in employee compensation.

Histogram for Salary:



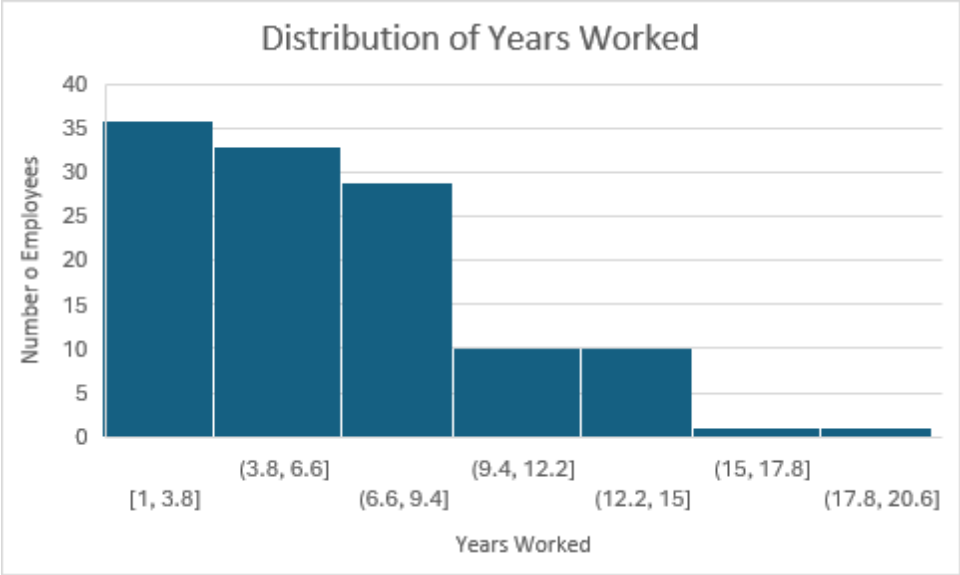
Histogram showing the distribution of employee salaries

Histogram for Age: Symmetrical; most employees are between ages 35 and 60. The distribution of ages is often symmetrical. With an average age of 47 and a median age of 49, most of the employees are in the 35–60 age range. This indicates a stable age distribution and a well-established workforce.



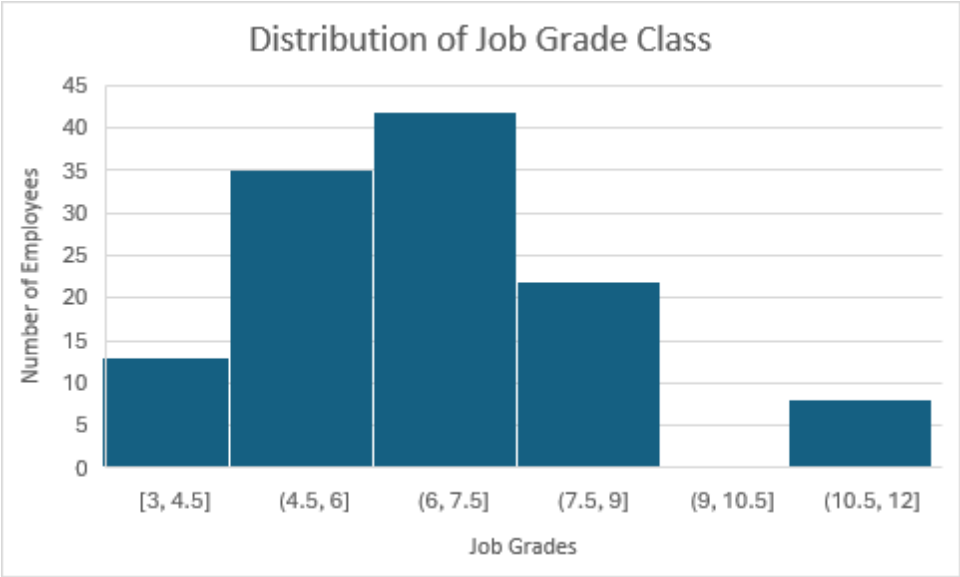
Histogram showing the distribution of employee ages

Histogram for Years Worked: Slightly right-skewed; most have 3–9 years of service. Many employees have worked for three to nine years, as indicated by the minor right-skewness in YearsWorked. On the upper end of the spectrum, just a small percentage of people have more than fifteen years of experience.



Histogram showing the distribution of years worked by employees

Histogram for Job Grade Classification: Most employees are in job grades 5–7, indicating a predominance of mid-level job roles. Fewer workers in both the lower and higher job classes result in a mild skew and a surge in mid-tier occupations. With a range of $31,200 to $82,300, the average salary is roughly $53,000.



Histogram showing the distribution of job grade classifications

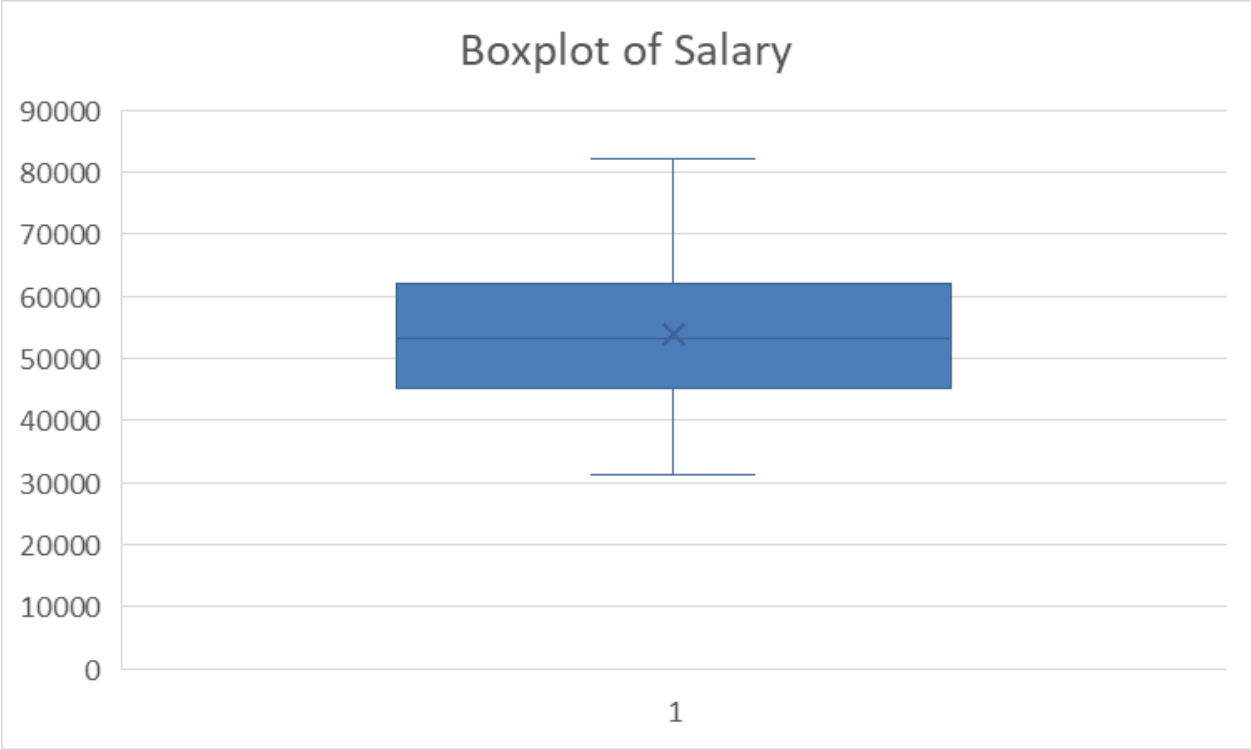
Years and AgeBoth are more symmetrical, according to work distributions. Years and AgeThere is a more equitable division of labor. The bulk of employees are between the ages of 35 and 60, however the age range is broad (19–74). The range is concentrated in the 47–49 age bracket. Years Worked has a slight right skew, with most employees having worked for three to nine years. A few outliers with over 15 years of expertise extend a small section of the distribution. Both lower and higher classes have fewer employees; grades 5 and 7 have the highest JGClass.

4. Outlier Analysis & Boxplots

To find possible outliers, boxplots were made for each numerical variable. A number of high-end outliers were visible in the salary boxplot, suggesting that some employees make far more than the norm. This could affect average salary calculations and regression analysis.

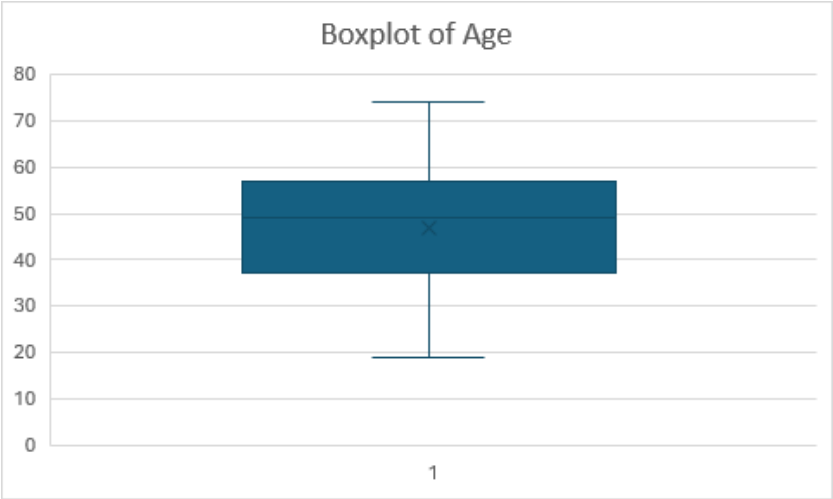
Salary: Several high-end outliers were displayed in the boxplot of salaries. These data indicates that some employees earn significantly more than others, may have an effect on regression estimations and the overall average. The majority of incomes fall into the middle level.

Boxplot for Salary:



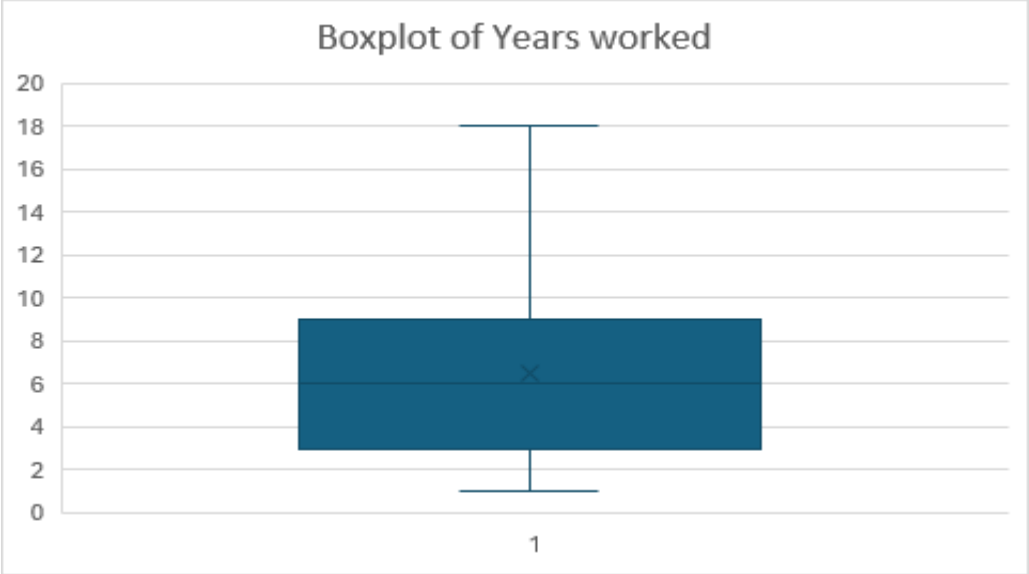
Boxplot of employee salaries showing outliers

Boxplot for Age: The age distribution appeared to be symmetrical overall, with a few small outliers on either end. The primary age trend among employees is not significantly affected by these outliers, despite the fact that they show slight variances.



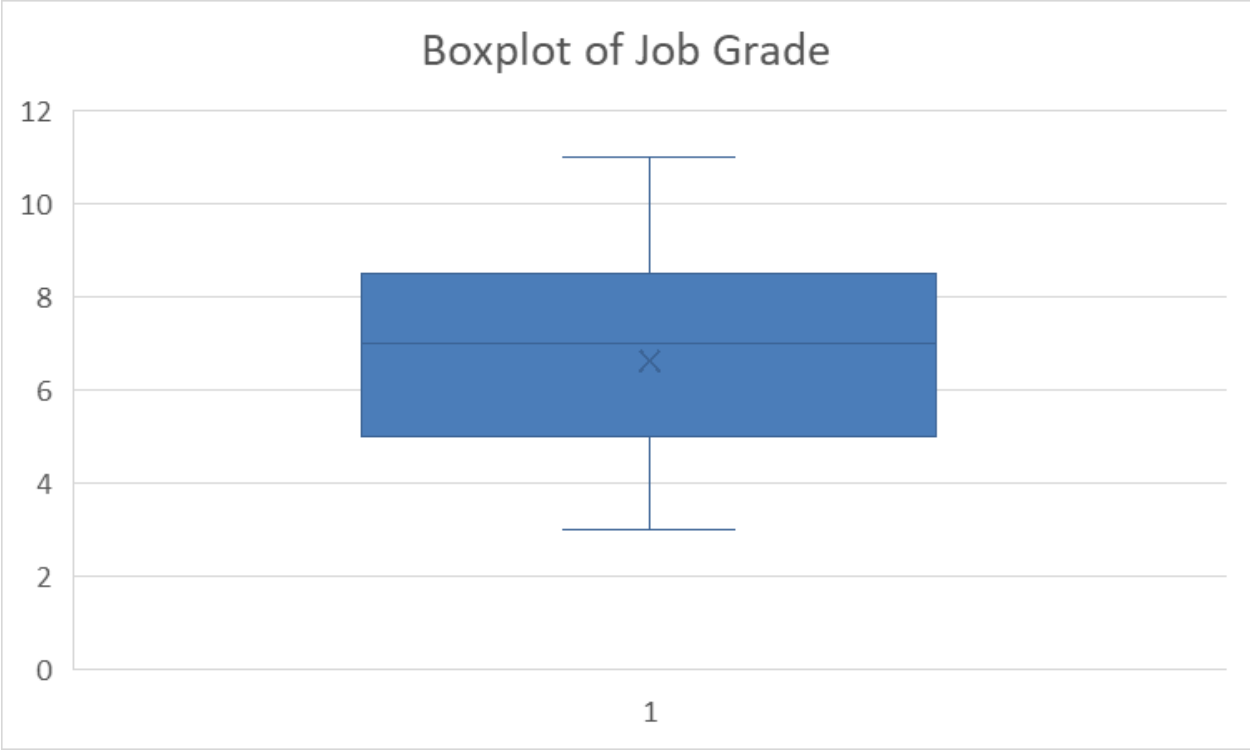
Boxplot of employee ages showing outliers

Boxplot for Years Worked: A few employees' very long tenures resulted in slight right-end outliers, but overall the Years Worked trend was symmetric. Most valuables are concentrated between three and nine years.



Boxplot of years worked showing outliers

Boxplot for JGClass: Since JGClass is a discrete variable, there were no obvious outliers and minimal fluctuation across levels. The distribution also supports mid-level job grades for employee concentration.



Boxplot of job grade classifications showing distribution

No extreme outliers were identified for JGClass. The job grade distribution is concentrated at mid-levels, reinforcing the histogram observations.

5. **Hypothesis Testing:** A question of interest for the dataset is - Is there any significant difference between the average salary of employees who are minorities compared to the ones who are non minority?

Null Hypothesis (Ho): Mean salary of minority group is equal to the mean salary of the non-minority group.

Alternate Hypothesis (Ha): Mean salaries of the two groups are different.

Group Statistics

	EthnicityCODE	N	Mean	Std. Deviation	Std. Error Mean
Salary	1	38	50097.37	11215.951	1819.468
	0	82	55678.05	11899.445	1314.074

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means							
						Significance				95% Confidence Interval of the Difference	
		F	Sig.	t	df	One-Sided p	Two-Sided p	Mean Difference	Std. Error Difference	Lower	Upper
Salary	Equal variances assumed	.088	.767	-2.433	118	.008	.016	-5580.680	2293.957	-10123.340	-1038.021
	Equal variances not assumed			-2.487	76.197	.008	.015	-5580.680	2244.382	-10050.569	-1110.792

Independent Samples Effect Sizes

				95% Confidence Interval	
		Standardizer[a]	Point Estimate	Lower	Upper
Salary	Cohen's d	11689.431	-.477	-.866	-.087
	Hedges' correction	11764.390	-.474	-.860	-.086
	Glass's delta	11899.445	-.469	-.859	-.076

a. The denominator used in estimating the effect sizes.
Cohen's d uses the pooled standard deviation.
Hedges' correction uses the pooled standard deviation, plus a correction factor.
Glass's delta uses the sample standard deviation of the control (i.e., the second) group.

Result: The average salary for employees who were not minorities(Ethnic Code 0) was $55,678.05 while the average salary for employees who were minorities(Ethnic Code 1) was about $50,097.37. SPSS performed two independent sample t-tests of equal means. We opted for the test of equal means for when Equal variances non-assumed. With a t-statistic of -2.487 and a p-value was 0.016, we reject the Ho. Therefore, we claim that there is a significant difference between the mean salaries of the minority and non-minority group.

This kind of conclusion may lead to the further company-wide data evaluation of the compensation policy to explorer whether there are any other variables that justify this disparity (years of service, level of education, level of specialty of job assignment, etc).

6. **Chi-square test of independence:** The two categorical data for performing a chi-square test of independence are JGClass and Ethnicity Label. They both

were used as categorical data to perform the chi-square test. To the question: Is there any correlation between the categorical version of Job Grade Classification (JGClass) and Ethnicity?

Null Hypothesis (Ho): Ethnicity and Job Grade Classification are independent categorical variables

Alternate Hypothesis (Ha): Ethnicity and Job Grade classification are correlated.

Case Processing Summary

	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
JGClass * Ethnicity_Label	120	100.0%	0	0.0%	120	100.0%

JGClass * Ethnicity_Label Crosstabulation

			Ethnicity_Label		
			Minority	Not Minority	Total
JGClass	3	Count	5	8	13
		Expected Count	4.1	8.9	13.0
	5	Count	7	28	35
		Expected Count	11.1	23.9	35.0
	7	Count	18	24	42
		Expected Count	13.3	28.7	42.0
	9	Count	5	17	22
		Expected Count	7.0	15.0	22.0
	11	Count	3	5	8
		Expected Count	2.5	5.5	8.0
Total		Count	38	82	120
		Expected Count	38.0	82.0	120.0

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	5.848[a]	4	.211
Likelihood Ratio	5.957	4	.202
N of Valid Cases	120		

a. 2 cells (20.0%) have expected count less than 5. The minimum expected count is 2.53.

Symmetric Measures

		Value	Approximate Significance
Nominal by Nominal	Phi	.221	.211
	Cramer's V	.221	.211
N of Valid Cases		120	

5.848 is the Chi-Square statistic with 4 degrees o freedom. The subsequent value is 0.211. Therefore, we fail to reject the null hypothesis at alpha=0.05 level of significance. We ae able to claim that there is no correlation between the two categorical variables of the event. Claim. Minority or Non-Minority status does not seem to have an any correlation with the work grade level of an employee. Both variables are independent.

7. **One way ANOVA:** ANOVA is a statistical test ideal to compare the Salary to the distinct categorical work grade levels, JGClass. That is the mean income across work grade levels. ANOVA will help us answer the question of Whether there is a pay differentiator across differing work grade levels?

Null Hypothesis (Ho) : The mean salary across all work grade levels is equal.

Alternate Hypothesis (Ha): The mean salary differs for at least one work grade level.

ANOVA Effect Sizes[a]

		Point Estimate	95% Confidence Interval Lower	Upper
Salary	Eta-squared	.691	.588	.747
	Epsilon-squared	.681	.573	.738
	Omega-squared Fixed-effect	.679	.571	.736
	Omega-squared Random-effect	.346	.250	.411

a. Eta-squared and Epsilon-squared are estimated based on the fixed-effect model.

Multiple Comparisons

Dependent Variable: Salary

Tukey HSD

(I) JGClass	(J) JGClass	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval Lower Bound	95% Confidence Interval Upper Bound
3	5	-7861.099*	2189.074	.004	-13928.19	-1794.00
	7	-16324.908*	2139.097	<.001	-22253.49	-10396.33
	9	-27256.294*	2357.743	<.001	-33790.86	-20721.73
	11	-36340.385*	3028.576	<.001	-44734.19	-27946.58
5	3	7861.099*	2189.074	.004	1794.00	13928.19
	7	-8463.810*	1542.525	<.001	-12738.97	-4188.65
	9	-19395.195*	1833.739	<.001	-24477.47	-14312.92
	11	-28479.286*	2641.197	<.001	-35799.45	-21159.12
7	3	16324.908*	2139.097	<.001	10396.33	22253.49
	5	8463.810*	1542.525	<.001	4188.65	12738.97
	9	-10931.385*	1773.779	<.001	-15847.48	-6015.29
	11	-20015.476*	2599.925	<.001	-27221.26	-12809.69
9	3	27256.294*	2357.743	<.001	20721.73	33790.86
	5	19395.195*	1833.739	<.001	14312.92	24477.47
	7	10931.385*	1773.779	<.001	6015.29	15847.48
	11	-9084.091*	2782.594	.012	-16796.15	-1372.04
11	3	36340.385*	3028.576	<.001	27946.58	44734.19
	5	28479.286*	2641.197	<.001	21159.12	35799.45
	7	20015.476*	2599.925	<.001	12809.69	27221.26
	9	9084.091*	2782.594	.012	1372.04	16796.15

Results: The F-statistic of 64.44 along with a p-value of less than 0.0001 are the result of an one-way ANOVA test. We reject the null hypothesis. We are able to claim that at the 5% significance level, the average salary is impacted by the different work grade levels. The is a correlation between The correlation between work grade levels and wages is strongly supported by this data.

8. **Summarizing the exploratory data analysis steps:** As a consequence of this research, we realize that is there is a significant association between

ethnicity status and salary. Minority employees make significantly less money than their non-minority peers, according to the results of the two-sample hypothesis t-test. The p-value of 0.016, indicates a significant disparity that cannot be explained by chance only, opening the door for follow-up studies to explore for other reasons. Next, the Chi-Square categorical factor test showed that there is no significant correlation between work grade level and ethnicity. There seems to be a pretty equal distribution of ethnically diverse workers across the different work grade levels.

Last, the one-way ANOVA showed that there is a strong correlation between work grade level and salary. Higher work grade level workers make more money, compared to lower work grade level workers. The statistical significance of this difference is shown by p-value (< 0.0001).

As a conclusion, we may claim that there are disparities in salary not sufficiently explained by work grade level alone. This suggests the need to audit company-wide policy and practices regarding hiring and compensation procedures.

Multiple Regression Model

The datasets consist of 120 observations and 6 variables. The variables provided information about demographics of the employees. These variables are Salary, Age, years at work, job grade class, ethnicity Code, and Ethnicity Label. Table 1 provides information about the nature of the data and variable types.

Variable	**Nature of Data**	**Type of Variable**	**Justification**
Salary	Numerical	Continuous	Can take on any value within a given range. It has a true zero point (no salary).
Age	Numerical	Continuous	Can take on any value within a range It has a true zero point (birth).
YrsWork	Numerical	Discrete	Counted in whole numbers (years). It has a true zero point (no work experience).
JGClass	Numerical	Ordinal	The values 1, 3, 5, 7, 9 and11 represent ordered categories or ranks. The difference between levels is not necessarily uniform or mathematically meaningful, but the order is important.
EthnicityCODE	Numerical	Categorical (Binary)	The numbers 0 and 1 are arbitrary labels used only to distinguish between two distinct, unordered groups. Mathematical operations are meaningless.
Ethnicity_Label	Text/String	Categorical (Nominal/Binary)	The labels Minority and Not Minority are used to distinguish between two distinct, unordered groups.

Table 1: Variables Types

1. **The multiple linear regression that was fitted to the data was:**

$$S = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4$$

Where:

S: represent the salary

x_1: represents the Age,

x_2: represents the Years of work,

x_3: represents the Job grade class,

x_4: represent the Ethnicity, and the β_i's are the regression coefficients

The least squares method was used to estimate the regression coefficients in R and the results of the fitted model are displayed in Figure 1.

```
Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)    38169.56    2497.02  15.286  < 2e-16 ***
Age               48.95      53.74   0.911  0.36429
YrsWork          296.72     172.31   1.722  0.08783 .
JGClass5        5500.15    2064.54   2.664  0.00886 **
JGClass7       15215.03    2052.52   7.413 2.52e-11 ***
JGClass9       23719.09    2454.79   9.662  < 2e-16 ***
JGClass11      32354.07    3216.92  10.057  < 2e-16 ***
EthnicityCODE1 -6041.67    1255.56  -4.812 4.70e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 6041 on 112 degrees of freedom
Multiple R-squared:  0.7586,    Adjusted R-squared:  0.7436
F-statistic: 50.29 on 7 and 112 DF,  p-value: < 2.2e-16
```

Figure 1: Regression model 1 Results

Thus, the fitted model is:

$$S = 38169.56 + 48.95x_1 + 296.72x_2 + 5500.15x_{3\ class\ 5} + 15215.03x_{3\ class\ 7} + 23719.09x_{3\ class\ 9} +$$

Results from Figure 1 shows that the fitted model is statistically significant. This is because the F-value= 50.29 on 7 and 112 degrees of freedom has a p-value of

$2.2e - 16$, which is less than 0.05. This is a result from the Anova test which has the hypothesis

H_0: $all\ of\ the\ coefficients\ \ are\ equal\ to\ zero$ vs

H_1: $atleast\ one\ \ of\ the\ coefficients\ is\ different\ from\ zero$

Since the p-value is less than 0.05, the null hypothesis is rejected and a conclusion that the model is statistically significant is made. Additionally, the R^2 of 0.7586 tells us that about 75.86% variation observed on the salary variable is explained by the model's independent variables, suggesting a strong fit. The variables years of work, job grade class and ethnicity are all statistically significant to the model, while the variable age was found to be statistically insignificant.

2.1 Backward Elimination Process

Results from Figure 1 showed that the variable age was statistically insignificant, hence it was removed and a new model was created without this variable. The results of the new model are presented in Figure 2.

Figure 2: Regression model 2 Results

Thus, the fitted model is:

$$S = 39724.5 + 344.3x_2 + 5928x_{3\ class\ 5} + 15777.9x_{3\ class\ 7} + 24557.8x_{3\ class\ 9} + 33124.3x_{3\ class\ 11}$$

Results from the Figure 2 shows that the fitted model is statistically significant. This is because the F-value= 58.63 on 6 and 114 degrees of freedom has a p-value of $2.2e-16$, which is less than 0.05. This is a result from the Anova test which has the hypothesis

H_0: $all\ of\ the\ coefficients\ \ are\ equal\ to\ zero$ vs

H_1: $atleast\ one\ \ of\ the\ coefficients\ is\ different\ from\ zero$

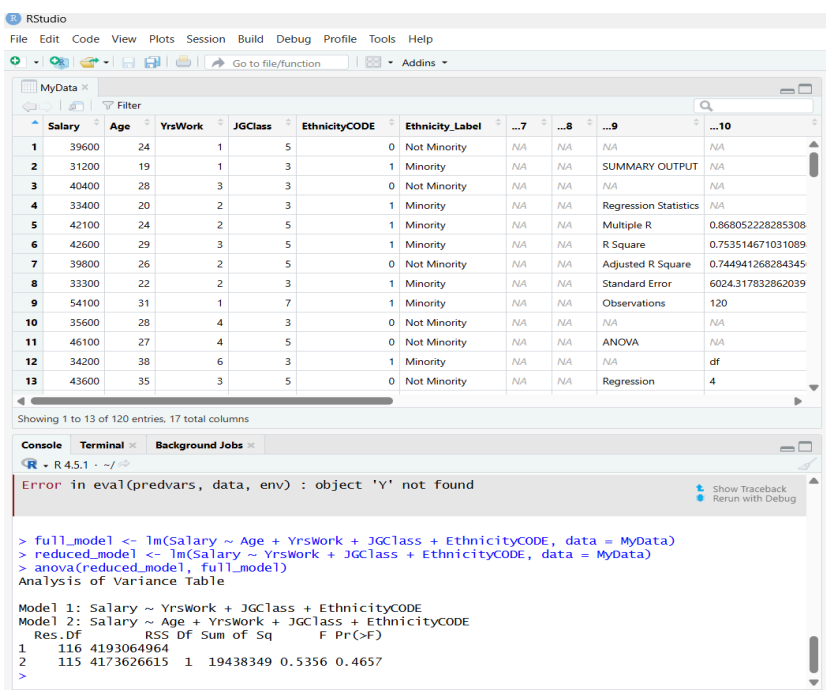
2.2 Comparing Model 1 and 2

Model	**R²**	**Adjusted R²**	**Notes**
1	0.7586	0.7436	Includes all variables (Age + YrsWork + JGClass + EthnicityCODE)
2	0.7569	0.7439	Removed Age (only YrsWork + JGClass + EthnicityCODE)

Since the p-value is less than 0.05, the null hypothesis is rejected and a conclusion that the model is statistically significant is made. Additionally, the model 2 R^2 of 0.7569 tells us that about 75.69% variation observed on the salary variable is explained by the model's independent variables, suggesting a strong fit.

2.3 Comment on behaviour of R(squared)

The value of R^2 in model 2 (0.7569) is slightly lower than that of model 1 which was (0.7586). This shows that model 1 was slightly better at accounting the variations in salary than model 2. However, the adjusted R^2 tells a different story with model 2 having a slightly higher R^2 (0.7439) versus (0.7436) for model 1. This difference stems from the fact that the adjusted R^2 value penalises the model for using useless predictors. Hence a conclusion that model 2 is better over model 1 is made and the decision to remove age was a good step for parsimony.



3. F Test Comparison of Model A & Model B

F-Statistic : 0.5356

P-Value : 0.4657

<u>Interpretation</u>

Since the P-Value (0.4657) is lower than 0.05 we fail to reject the null hypothesis meaning that the removal of the age variable does not make the model significantly worse.

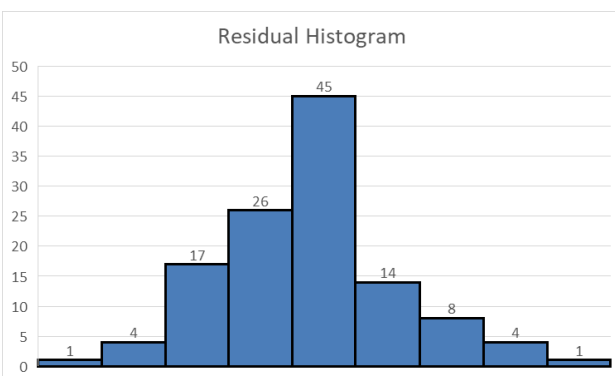
4. Final Model Residual Analysis

In this stage we will analyze the residuals that were obtained from Model B, our final regression model. We will assess how appropriate the model is through evaluating the Normality, Equal/Constant Variance assumptions and the Independence through the residual plots.

Assumptions and Analysis

(i) **Normality Assumptions**

Shown below are the residual plots obtained from Model B, our final regression model



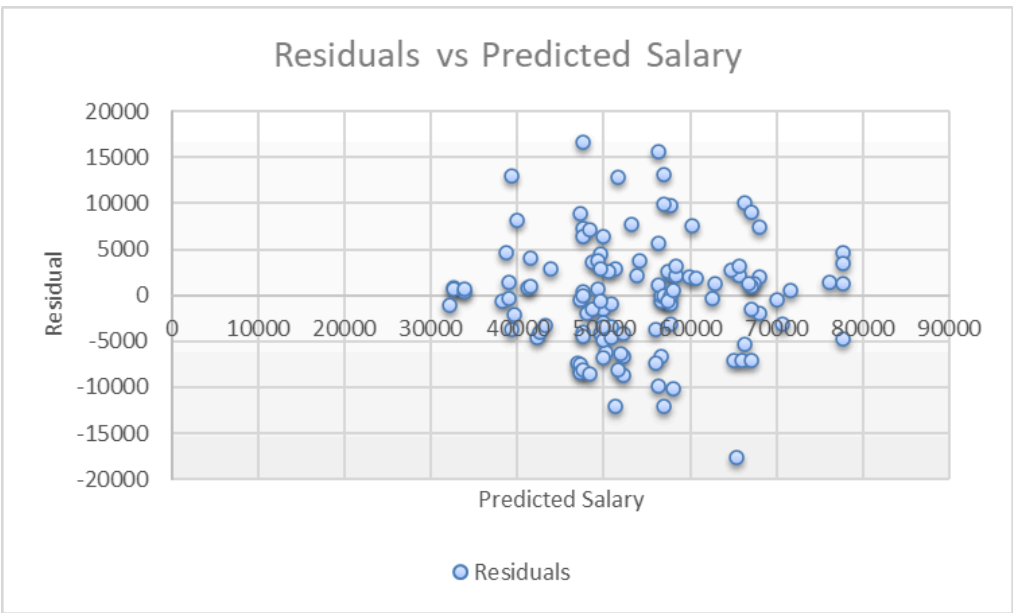
The residual histogram is almost bell shaped with approximate symmetry, however it is not perfectly normal with a slight skewness and deviations within the results that lean towards extreme values.

(1) Evidence: The residuals are randomly scattered around zero with extremes on either side according to the histogram.

(2) Violation: There is little violation of normality due to skewness.

(3) Possible Remedy: A log transformation can be applied to Salary in order to increase normality.

(ii) **Independence Assumption**



The residuals vs predicted salary scatter plot does not show any discernable pattern.

(1) Evidence: The predicted values do not contain any noticeable trends nor do they follow any systematic pattern.

(2) Violation: No Violation

(3) Homoscedasticity: The Equal/Constant Variance Assumption is satisfied.

(iii) Equal/Constant Variance Assumption (Homoscedasticity)

After analyzing the scatter plot we can see that the residuals are similarly spread out among the salary values within the middle. However as the salary increases, so does the spread of the residuals.

(1) Evidence: Low salary residuals are more dense compared to the high salary residuals.

(2) Violation: There is partial violation of heteroscedasticity with the funneling of residuals

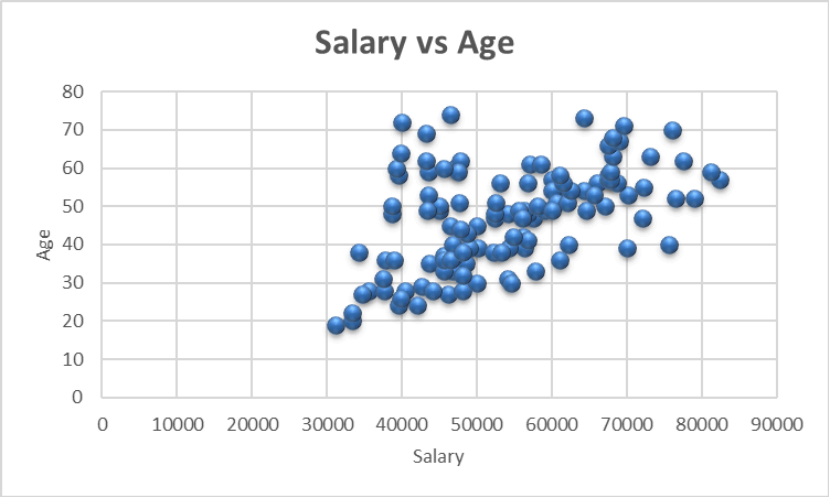
(3) Remedy: Using Weighted Least Squares (WLS) regression can fix the issue with heteroscedasticity.

Residual analysis of Model B indicates the assumptions are generally satisfied, however the slight normality and constant variance should be noted. While the effects will not have a significant impact on the model's validity, we can use variable transformation or weighted least squares to refine the results.

5. Linearity Assumption Analysis

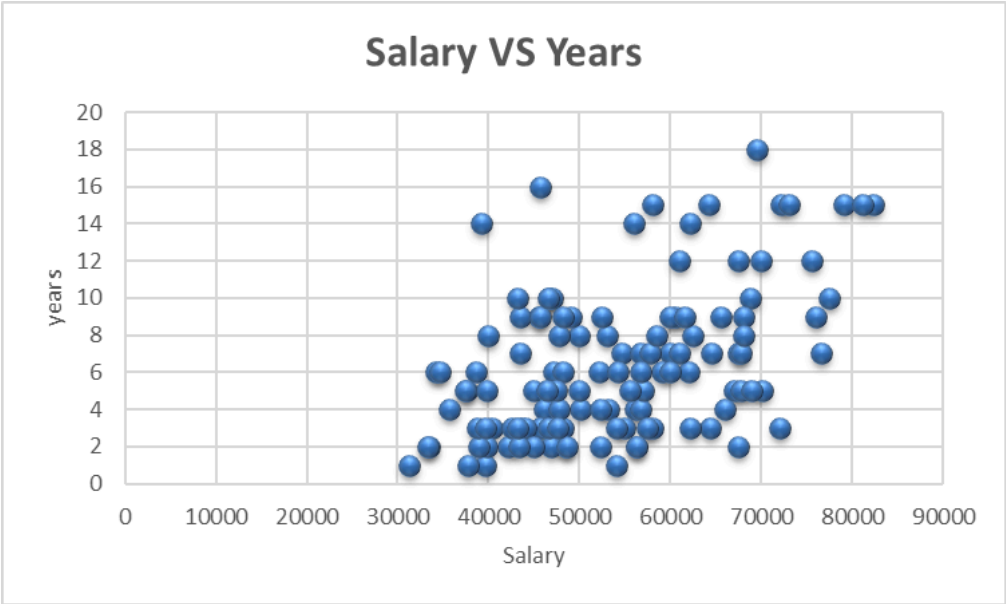
We will use scatter plots to evaluate the linearity assumption, by analyzing each scatterplot created between each independent variable and the dependent variable (Salary). The observed patterns within each scatter plot will be interpreted in this section.

(i) <u>Salary VS Age</u>



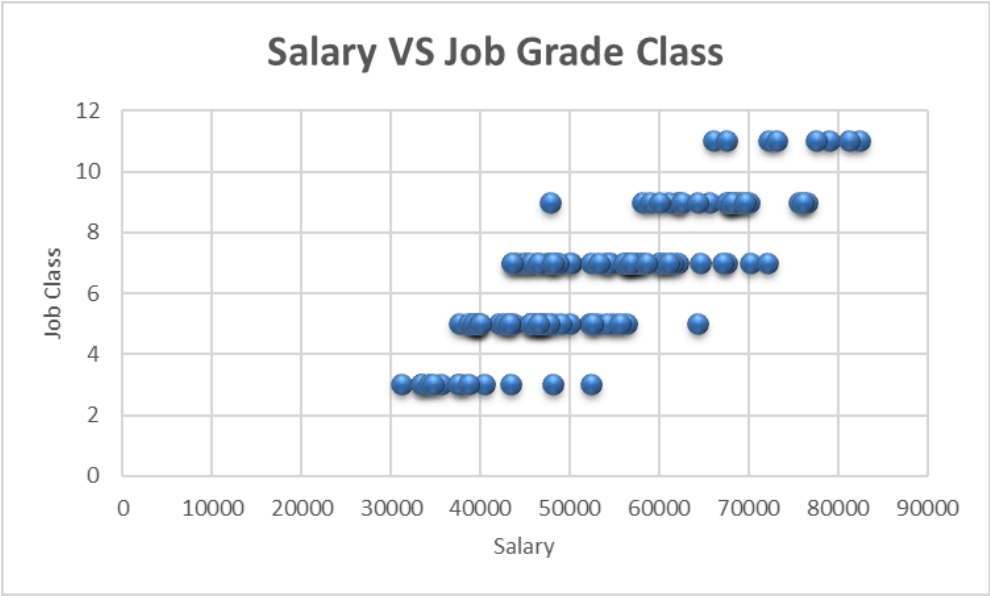
The scatter plot is in a mostly linear positive relationship between salary and age, there is noticeable dispersion which can be due to minor deviations however, there is not a significant violation of the linearity assumption with no transformation required. Overall as age increases the salary will increase as well.

(ii) Salary VS Years Worked



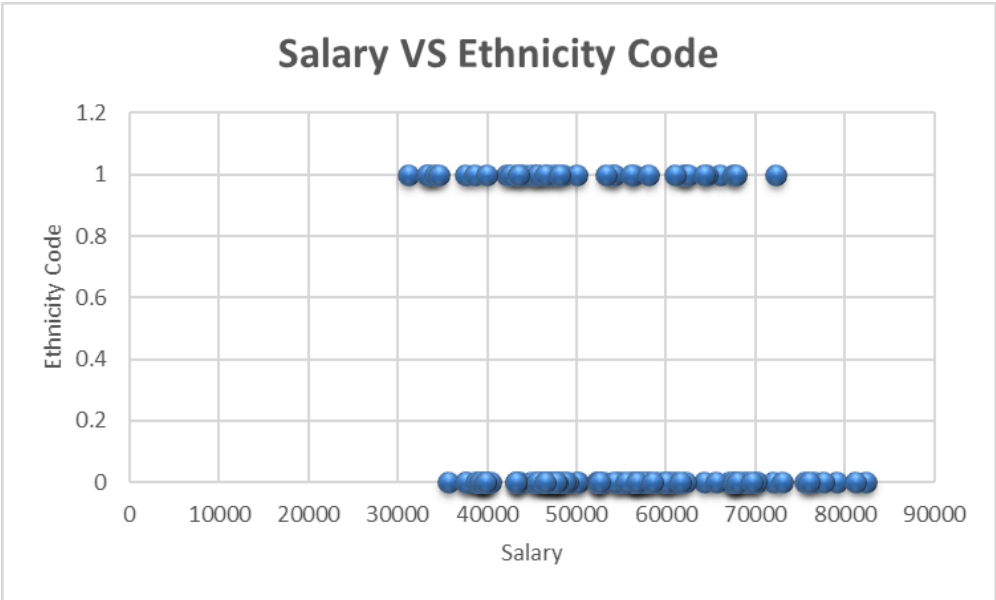
The scatter plot indicates a positive linear relationship between salary and years worked. The data points vary slightly in dispersion but still trend upward without any violations observed nor any need for transformations. Ultimately as years worked increases the salary will increase as well.

(iii) Salary VS Job Grade Class



The scatterplot takes a stepwise pattern due to the fact that Job Grade Class has ordinal variables consisting of odd numbers between 3-11. The salary increases along with the job grade class but since the results contain an ordinal variable we cannot establish a linear relationship. No transformation is required

(iv) Salary VS Ethnicity Code



The scatter plot carries two independent groups that are identified under two categories within the ethnicity code; 0 = Not a Minority and 1 = Minority). The ethnicity code is a binary categorical variable which means we cannot establish a linear relationship. Thus we cannot apply the linearity assumption and we do not need to apply a transformation.

We can conclude that the linearity assumption is reasonably satisfied for the continuous variables given the scatter plots above (Years Worked and Age). There were not any major violations detected and no transformations were required. We cannot apply the linearity

assumption to Job Grade Class and Ethnicity Code due to the simplistic nature of their values. Given these results no corrective actions are necessary.

6. Final Regression Model and Interpretation

A. Final Model Equation (to 2 decimal places):

Y = 23836.64+(38.66*Age)+ (291.61*YrsWork) + (4265.49*JGClass)-(5847.88*EthnicityCode)

Variables:

Y = Predicted Salary

YrsWork = Years Worked

Age = Age in Years

JGClass= Job Grade Class

EthnicityCODE: 0 = Not a Minority, 1 = Minority

B. Interpretation of Coefficients:

(i) Intercept (23836.64): The expected salary for a non-minority that has 0 years worked, age is 0, and a Job Grade Class of 0.

(ii) Age (38.66): Every additional year in age the salary will increase by $38.66 (not statistically significant).

(iii) Years Worked (291.61): Every additional year worked the salary will increase by $291.61.

(iv) Job Grade Class (4265.49): For every increase in job class the salary will increase by $4265.49.

(v) Ethnicity Code (-5847.88): Minority Employees earn $5847.88 less than non minorities with all else being constant.

C. Coefficient of Determination (R-Squared)

The R-Squared value is 0.7535 which means that 75.35% of the variability in salary can be explained by this model. The adjusted R-Squared is 0.7449 which would mean that all of the variables make a significant contribution to the model due to a very minor difference between R-Squared and adjusted R-Squared. The fit of this model is strong for the human resources dataset.

D. Standard Error of the Estimate (SYX):

The standard error of the estimate shows the amount that salary actually deviates from the predicted salary on average. We can determine that on average our predicted salary differs by $6024.32 from our actual salary. With an error of about 11 percent we can determine that our model has good predictive power.

E. Example Prediction

Y = 23836.64+(38.66*6)+ (291.61*8) + (4265.49*3)- (5847.88*1) =33350.07

Age = 6

YrsWork= 8

JGClass=3

EthnicityCode=1

F. 95% Confidence Interval

Our narrow range for our mean salary estimate of individuals with similar characteristics

33,350.07±1,099.6 = ($32,250, $34,450)

G. 95% Prediction Interval

Our wider range for our salary estimate while taking into account variability in salaries

33,350.07±12,048.64 =($21,301, $45,399)

H. Comment

We are 95% confident that an individual that has the characteristics listed in E will have a salary between $32,250 and $34,450. We can also say with the same confidence that the actual salary of one individual with the characteristics found in section E will have a salary between ($21,301,$45,399). Our prediction interval can tell what a particular person will make given these variables while our confidence interval tells us what the average person will make via salary.

Conclusion:

Exploratory data analysis and multiple regression modeling were used in the study of the company's salary structure. The information from 120 employees was used to identify trends and pay inequalities through analysis tools such as regression, ANOVA, t-tests and chi-square tests. The tests demonstrated a continuous pay

disparity between minorities and non minorities, with the latter making a larger salary. The chi squared test did not reveal any significant correlation between ethnicity and work level so the resulting explanation would have to be due to job grade. We verified that pay will rise hand in hand with job grade through the use of ANOVA. Ultimately even with the data showing equal promotion opportunities, there is still a significant difference in salary between minorities and non minorities.

This will have a major impact on the organization's equity practices. Although there are opportunities for fair promotions, the existing pay disparities must be addressed promptly. Reviewing compensation policies, entry-level salaries, and pay raise procedures can help ensure that unconscious bias does not influence salary decisions. Ultimately, these findings emphasize the need for data-driven HR policies to guide future efforts toward inclusive practices and equitable compensation.